Does LLM Loneliness Detection Align with Expert Evidence?

Anonymous Author(s)

Abstract

In this paper, we show that accurate loneliness detection does not guarantee that a model relies on evidence that experts consider meaningful. To examine this distinction, we evaluate loneliness models along three dimensions: *detection*, whether they predict loneliness correctly; *plausibility*, whether their attributed evidence aligns with expert annotations; and *faithfulness*, whether that evidence affects the model's predictions. We compare full fine-tuning, probing, and our proposed CLEAN-FT, which first locates loneliness with an attention-pooled linear probe at an intermediate layer of the frozen model, then fine-tunes only the layers beneath that probe so the decision stays where the concept is already represented. On the public LonXplain dataset, attention-pooled probing approaches full fine-tuning in detection while producing more plausible and faithful evidence. Full fine-tuning achieves strong detection but weaker expert alignment, whereas CLEAN-FT achieves the strongest detection while retaining stronger evidence alignment and faithfulness. On expert-annotated conversations between older adults and a voice chatbot, CLEAN-FT maintains stronger evidence alignment and faithfulness than full fine-tuning under distribution shift. Overall, training strategy shapes not only detection performance, but also whether a model's explanation reflects the evidence it relies on, and whether that alignment survives distribution shift.

Keywords: Loneliness detection; LLM; Interpretability; Concept Localization

Data and Code Availability The OOD conversation data contain private information and cannot be fully released; de-identified transcripts, expert labels, annotation instructions, and code will be released upon acceptance.

Institutional Review Board (IRB) The study was approved by the authors' Institutional Review Board, and all participants provided informed consent; full details will be provided upon acceptance.

1. Introduction

Loneliness is a major public health crisis, affecting 20% to 40% of older adults (Yamada et al., 2021; Cho et al., 2025) with a mortality risk comparable to smoking or obesity (Holt-Lunstad et al., 2015). Standard clinical tools like the UCLA (Russell, 1996) or De Jong Gierveld scales (De Jong-Gierveld and van Tilburg, 2006) compress complex, multi-week experiences into a single retrospective score and often trigger under-reporting due to stigma (Barreto et al., 2022). In contrast, people frequently express their social isolation through spontaneous language in everyday writing or conversation. Analyzing this naturalistic text offers a more candid, continuous reflection of social well-being that bypasses the limitations of formal questionnaires.

NLP research on loneliness has evolved from handcrafted features to contextual transformers and large language models. Early work relied on linguistic markers like pronoun density and syntax (Badal et al., 2021; Guntuku et al., 2019), with later methods pairing dictionary features with attention mechanisms (Wang et al., 2024). Modern architectures pushed predictive metrics even higher, with BERT achieving 97% accuracy on social media posts (Jiang et al., 2022) and RoBERTa reaching an F1 of 0.92 on clinical care records (Rickman et al., 2025). Most recently, LLM prompting has been used to score free text against clinical scales (Kim et al., 2026) and output open-ended rationales (Yang et al., 2023). Yet strong task performance alone fails to show whether these models' internal representations capture clinically meaningful evidence or simply exploit spurious shortcuts.

Answering this question requires evaluating more than whether a model predicts loneliness correctly. We therefore evaluate loneliness models along three dimensions: *detection*, whether they predict loneliness correctly; *plausibility*, whether model-highlighted words align with expert-annotated evidence; and *faithfulness*, whether removing those words changes the model's prediction (Jacovi and

© .

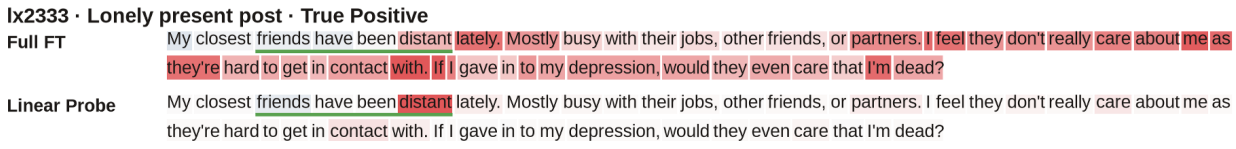


Figure 1: Per-token attribution (DLA) for full fine-tuning (Full FT) and the attention-pooled linear probe on a LonXplain true positive. Red pushes toward "lonely", blue away; green underlines mark expert-annotated spans.

Goldberg, 2020; DeYoung et al., 2020). This distinction matters because a model can make accurate predictions while highlighting evidence that does not reflect its actual decision process. We further examine whether these properties persist beyond Reddit by evaluating conversations between older adults and a chatbot under distribution shift.

The LonXplain dataset (Garg et al., 2023) provides whether a Reddit post is lonely or not (*post-label*), along with expert-annotated explanation on why the post is lonely (*span-label*). This makes it possible to go beyond prediction accuracy and compare what a model predicts, the evidence it highlights, and the words that actually affect its prediction. Figure 1 reveals a challenge: on the same true positive, both the fully fine-tuned model and the frozen linear probe predict loneliness correctly, but the probe concentrates its token-level attribution on the expert-annotated span while the fine-tuned model spreads attribution across the post.

We investigate this difference through *concept localization*: where loneliness is represented in the network and how training changes that representation. A frozen probe tests whether loneliness can be read out as a linear direction from an intermediate layer, where higher-level semantic information is often represented (Skean et al., 2025). Full fine-tuning instead updates both the backbone and classifier, which can distort pretrained features and encourage dataset-specific shortcuts (Kumar et al., 2022; Kirichenko et al., 2022). These changes may shift the model's decision away from the representation that originally encoded loneliness. We therefore compare probing, full fine-tuning, and hybrid training strategies across detection, evidence alignment, and faithfulness. We organize the study around three research questions:

RQ1 (Detection): Does probing perform as well as fine-tuning for loneliness detection, and how does pooling affect performance?

RQ2 (Evidence): Which training strategies produce evidence that is both aligned with expert annotations (*plausibility*) and faithful to the model's predictions (*faithfulness*)?

RQ3 (OOD Transfer): Do the differences in evidence alignment and faithfulness observed on LonXplain extend to conversations between chatbots and older adults under distribution shift?

Our contributions are:

- **Evaluation and finding.** We jointly evaluate detection, plausibility, and faithfulness across training strategies. Full fine-tuning detects loneliness well, but the evidence it highlights drifts from the words experts annotate. Probing and hybrid methods align highlighted evidence, model behavior, and expert annotation.

- **Method.** We propose CLEAN-FT (Concept-Localized Evidence-Aligned Fine-Tuning), which keeps the decision where the concept is represented. A linear probe locates loneliness in an intermediate layer, and fine-tuning then starts from that probe and adapts only the layers beneath it. CLEAN-FT achieves the strongest detection on both backbones while retaining stronger evidence alignment and faithfulness than full fine-tuning, with similar advantages under distribution shift.

- **Data and transfer.** We contribute an expert-annotated conversation dataset between a chatbot and older adults, then show that alignment and faithfulness patterns persist under distribution shift without retraining.

2. Method

Figure 2 summarizes the study in three stages. Stages 1 and 2 use LonXplain, comprising 3,522 Reddit posts labeled for loneliness, with expert-annotated evidence spans for a subset. Stage 1 uses post-level labels to evaluate detection, while Stage 2 evaluates which words the models identify as evidence. Stage 3 trans-

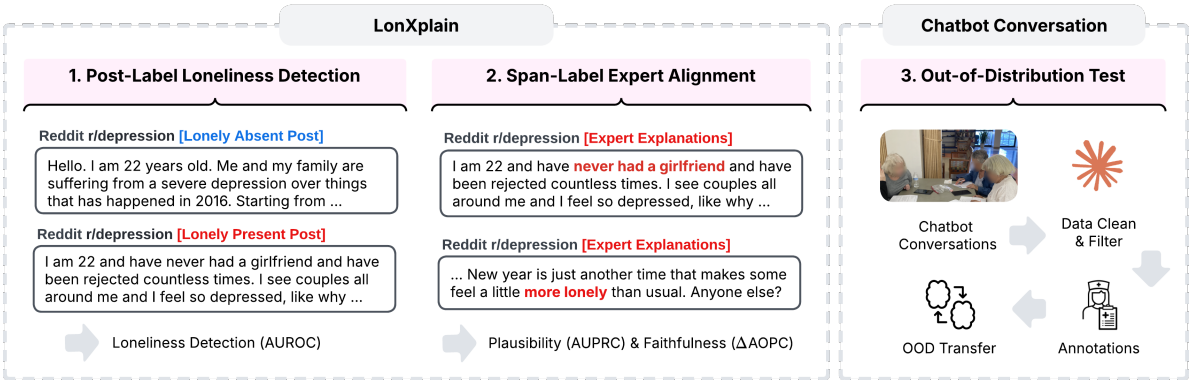


Figure 2: Three-stage design. Stage 1 trains all methods on LonXplain post labels and evaluates loneliness detection (AUROC). Stage 2 evaluates word-level evidence against expert spans for plausibility (AUPRC) and faithfulness (AOPC). Stage 3 tests the Stage 1 detectors out of distribution on older-adult chatbot conversations.

fers the Stage 1 detectors to chatbot conversations with older adults without retraining.

2.1. LonXplain Loneliness Detection

Loneliness Detection. We use Area Under the Receiver Operating Characteristic curve (AUROC) as the primary threshold-free metric:

$$\text{AUROC} = \mathbb{P}\big(f(x_+) > f(x_-)\big), \tag{1}$$

where $f(x)$ is the model's continuous loneliness score, and x_+ and x_- denote a lonely and non-lonely post, respectively. Thus, AUROC measures how often the model ranks a randomly selected lonely post above a non-lonely post. We compute AUROC from pooled out-of-fold predictions, allowing prompting, probing, and fine-tuning to be compared without a shared decision threshold or score scale. We do not use F_1 as the primary metric because it depends on a chosen threshold and class distribution.

2.2. LonXplain Expert Alignment

Word-Level Evidence Scores. We use two complementary word-level scores to distinguish *what the model's decision is composed of* from *which words the decision depends on*. Here, a word-level score assigns an importance value to each word in a post, allowing the words to be ranked from most to least important.

The first score is direct logit attribution (DLA), which decomposes the model's classification score into token-level contributions. The classification score is the value produced by the final classifier before converting it to a probability. For a linear classifier, the contribution of token t is

$$s_t^{\text{DLA}} = \alpha_t \, (w \cdot X_t), \tag{2}$$

where X_t is the token representation, w is the classifier weight vector, and α_t is the token's pooling weight. The contributions sum to the classification score, $f(x) = \sum_t s_t^{\text{DLA}} + b$, so DLA is an exact decomposition rather than an approximation. This makes it possible to inspect which words contribute to the model's prediction. Subword scores are mapped to words by taking the maximum score among each word's subwords, and we rank words by their DLA scores, denoted ρ_{DLA}.

The second score is occlusion, which measures how much the model's classification score changes when a word is removed:

$$s_i^{\text{occ}} = f(x) - f(x_{\setminus i}), \tag{3}$$

where $x_{\setminus i}$ is the post with word i removed. Occlusion is computed per word, and we rank words by these scores as ρ_{occ}. Thus, DLA asks which words contribute to the model's computed decision, whereas occlusion asks which words the decision depends on when they are removed.

Plausibility. Plausibility asks whether a method identifies the same words that experts regard as evidence for loneliness. We therefore compare each word ranking ρ with the expert-annotated spans using word-level AUPRC (Jacovi and Goldberg, 2020; DeYoung et al., 2020). Words overlapping an expert

span are treated as positives:

$$\mathrm{AUPRC}(\rho) = \sum_{k=1}^{n} (R_k - R_{k-1})\, P_k, \qquad R_0 = 0, \quad (4)$$

where n is the number of scored words in the post and P_k and R_k are precision and recall among the top-k words in ranking ρ. We report $\mathrm{AUPRC}(\rho_{\mathrm{DLA}})$ and $\mathrm{AUPRC}(\rho_{\mathrm{occ}})$.

The two rankings answer different questions but are evaluated against the same human evidence. DLA tests whether the model's own word-level explanation highlights expert-identified evidence. Occlusion tests whether the words that actually affect the model's score are the words experts identify as evidence.

Faithfulness. Faithfulness asks whether an explanation identifies words that actually affect the prediction. DLA is an explanation of the model's decision, so its faithfulness must be tested. Occlusion directly measures the decision's dependence on each word, so it does not require a separate faithfulness test. We therefore evaluate only DLA using deletion-based perturbation (DeYoung et al., 2020), summarized by the area over the perturbation curve (AOPC) (Samek et al., 2016). For each post, we progressively remove the highest-ranked positively scored words under ρ_{DLA} and measure the resulting decrease in the model's classification score:

$$\mathrm{AOPC}(\rho) = \frac{1}{50} \int_0^{50} \frac{f(x) - f\big(x_{\setminus p}^{(\rho)}\big)}{f(x) - f(\emptyset)}\, dp, \qquad (5)$$

where $x_{\setminus p}^{(\rho)}$ is the post after removing the top $p\%$ of its positively scored words under ranking ρ (at least one word), and $f(\emptyset)$ is the score for an empty post. We evaluate $p \in \{1, 2, 5, 10, 20, 50\}$ and approximate the integral using the trapezoid rule.

Because removing words can lower the score even when the ranking is uninformative, we report the improvement over a count-matched random ranking:

$$\Delta\mathrm{AOPC}(\rho) = \mathrm{AOPC}(\rho) - \frac{1}{5} \sum_{j=1}^{5} \mathrm{AOPC}(\pi_j), \qquad (6)$$

where each π_j is a random ranking over all scored words that removes the same number of words at each p. A positive ΔAOPC therefore indicates that the ranked words affect the prediction more than randomly selected words. Posts for which $f(x) - f(\emptyset) \leq 0$, or whose magnitude is below 0.1, are excluded because the normalization is undefined or unstable.

2.3. Chatbot Conversation OOD Test

Data. The out-of-distribution corpus consists of voice-based chatbot sessions with older adults: 16 participants, each contributing at least four weeks of daily sessions (383 in total) collected between December 2025 and August 2026. Collection details are provided in Appendix A.

Cleaning and Filtering. We clean the transcribed conversation by removing filler sounds, stutters, and recognized artifacts. Then, Claude Opus 5 (Anthropic, 2026) screens 383 sessions to filter conversations that appear to express loneliness, resulting in 34 sessions from 7 participants for expert annotation. To minimize false negatives, we apply a broad screening prompt (see Appendix B) to capture potential instances of loneliness.

Expert Annotation. Two expert annotators, each with extensive clinical experience caring for older adults experiencing loneliness and expertise in qualitative analysis, independently reviewed the 34 flagged sessions. They labeled whether loneliness was expressed and marked supporting evidence following the LonXplain definition of lonesomeness (Garg et al., 2023) (annotation rubric in Appendix C). We retain both annotations independently without resolving disagreements. We report percent agreement for labels and word-level Jaccard overlap (Jaccard, 1912) between the two annotators.

Out-of-Distribution Test. We transfer each Stage 1 detector to the older-adult conversations without retraining or threshold tuning and evaluate AUPRC and ΔAOPC using the same protocol as Stage 2.

3. Experiment

Shared setup. All LLM-based methods use Gemma-2-2B as the primary backbone and Llama-3.1-8B as a control. Each post is wrapped in the backbone's chat template, with template tokens excluded from pooling. Frozen probes and mid-layer methods read from a late-intermediate layer fixed a priori and never tuned on labels: Layer 17 for Gemma-2-2B (0.65 relative depth) and Layer 24 for Llama-3.1-8B (0.75). We use different relative depths because useful abstract concepts can emerge at different points across architectures (Meng et al., 2022). Figure 3 shows the three training families: fine-tuning, probing, and hybrid (proposed) methods. Each family is

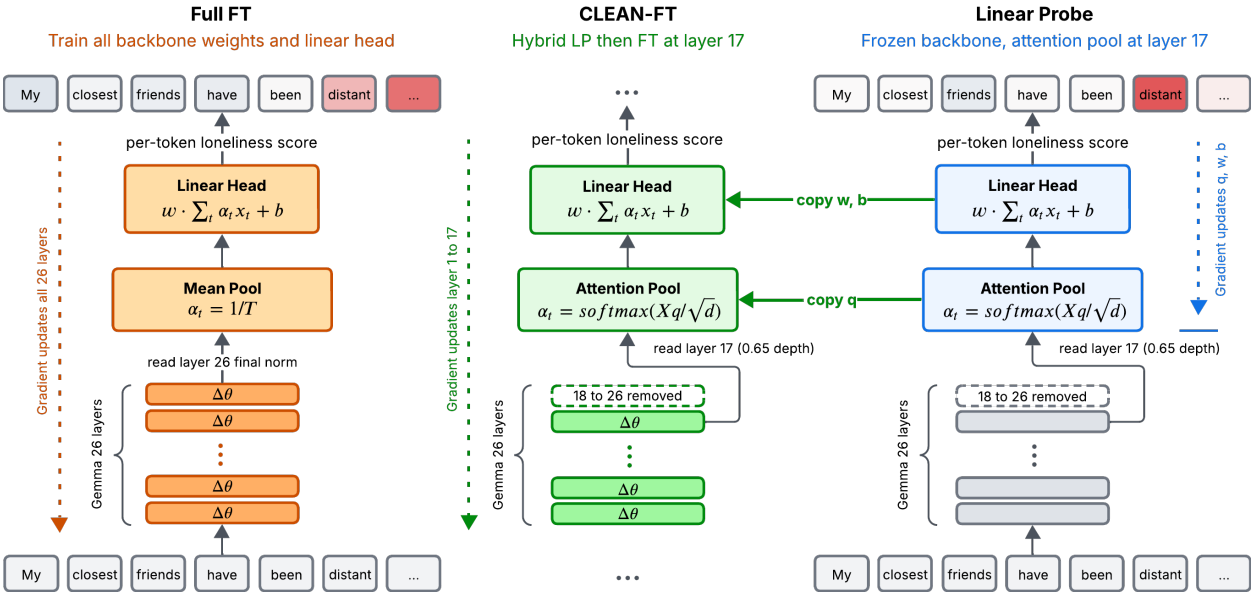


Figure 3: The three training methods on Gemma-2-2B. Full FT is shown with mean pooling, while CLEAN-FT and Linear Probe are shown with attention pooling for illustration; all methods are evaluated under both pooling strategies.

tested with two pooling strategies, differing only in how token representations are aggregated.

Evaluation Set. We divide the posts into ten stratified folds using a fixed random seed. Five designated folds $(0, 2, 4, 6, 8)$ form a shared test set of $1,761$ posts. For each fold, models are trained on the remaining nine folds ($3,170$ posts) and evaluated on the held-out fold. We use this shared test set for both loneliness detection and word-level evidence evaluation. For detection, models are evaluated on all held-out posts. For evidence evaluation, we use up to 100 held-out positive posts per fold containing expert-annotated spans, yielding $N = 499$ posts. The same 499 posts are used for all plausibility and faithfulness analyses.

3.1. Fine-Tuning Methods

Full Fine-Tuning (Full FT). Full FT updates every backbone parameter together with a randomly initialized linear head on the final layer. All fine-tuning methods share one optimization recipe: 8-bit AdamW with binary cross-entropy loss.

ML-FT. Mid-layer fine-tuning (ML-FT) places the classification head at the fixed intermediate layer, motivated by evidence that intermediate layers encode higher-level concepts before later layers specialize toward token prediction (Skean et al., 2025). It discards the layers above this point and fine-tunes the remaining layers using the Full FT recipe, with a randomly initialized head and no probe. Because ML-FT and the frozen probes read the same layer, their comparison isolates the effect of fine-tuning at a fixed depth, while comparison with Full FT isolates the effect of read-out depth.

3.2. Probing Methods

Probes train lightweight classifiers on hidden states of the frozen backbone (Belinkov, 2022). The backbone never changes so a probe can only detect loneliness if the pretrained representation already encodes it.

Linear Probe. An L2-regularized logistic regression on the residual representation, the simplest reference for what the frozen model supports.

MLP Probe. The linear head is replaced by a one-hidden-layer ReLU network with 256 units, testing whether a nonlinear read-out extracts loneliness information a linear probe cannot. Since an MLP has no single weight vector, its word-level evidence uses the Integrated Gradients direction of the MLP in place of w in Eq. 2 (Sundararajan et al., 2017); the shares still sum to the score, and coincide with the linear case when the MLP is linear.

SAE Probe. The residual representation is mapped to sparse, non-negative latent features by a pretrained sparse autoencoder (Huben et al., 2024): Gemma Scope at layer 17 for Gemma (Lieberum et al., 2024) and Llama Scope at layer 23 for Llama (He et al., 2024). An L1-regularized linear head tests whether this learned feature basis improves detection over the raw residual stream. Word-level evidence uses the same decomposition over the token-level latents. We evaluate the SAE probe with mean pooling only, since learning an attention query over 32,768 latents with only ∼3,000 posts would be poorly constrained and make the attention-pooled result difficult to interpret.

3.3. Hybrid Methods

LP-FT. Linear probe then fine-tuning follows Kumar et al. (2022): we first solve a linear probe on the frozen final-layer representation, then unfreeze the backbone and fine-tune from the probe with the Full FT recipe. LP-FT therefore provides representational localization through probe initialization, while retaining the final-layer read-out.

CLEAN-FT (proposed). CLEAN-FT (Concept-Localized Evidence-Aligned Fine-Tuning) combines structural and representational localization. Structurally, the classification head is placed at the fixed intermediate layer and the layers above it are discarded, so the decision is read directly from where loneliness is linearly decodable. Representationally, the head is initialized from a linear probe at that layer, so fine-tuning starts from an existing concept-aligned direction. In contrast, Full FT uses a random head at the final layer, LP-FT provides only representational localization, and ML-FT provides only structural localization. Comparing LP-FT with CLEAN-FT therefore tests the structural component, while comparing ML-FT with CLEAN-FT tests the representational component.

3.4. Reference Baselines

Span-FT (upper reference). All other methods learn from post labels alone. Span-FT extends Full FT by additionally training on expert-annotated span labels, providing direct supervision for which words should be identified as evidence. It serves as an upper reference for expert alignment, showing how well word-level evidence can be recovered when span annotations are available during training.

MentalBERT. MentalBERT (Ji et al., 2022) is a BERT-base encoder pretrained on mental-health forum text. We fine-tune all of its parameters with a linear head over mean-pooled tokens under a standard BERT recipe, as a domain-specific encoder baseline.

GloVe + BiGRU. Following LonXplain (Garg et al., 2023), frozen 300-d GloVe embeddings feed a bidirectional GRU with mean pooling and a linear head, a recurrent baseline that isolates the contribution of the LLM backbone.

4. Results

RQ1: Attention-pooled probing detects loneliness nearly as well as fine-tuning. Table 1 reports AUROC on the shared test set with 95% bootstrap half-widths. Under attention pooling, CLEAN-FT is the strongest detector on both backbones, and the frozen linear probe matches full fine-tuning on Gemma and trails it by only 0.015 on Llama. All decoder methods outperform GloVe+BiGRU, MentalBERT sits at the level of Full FT on Llama, and Span-FT performs no better than post-level training, so span supervision is unnecessary for detection.

The Δ column shows where pooling matters. Attention pooling adds 0.07 to 0.09 AUROC for the linear probe and 0.04 to 0.07 for the hybrid methods, but almost nothing for full fine-tuning. With a frozen backbone, mean pooling dilutes a sparse loneliness signal across neutral tokens, while a learned attention query recovers it. The relative insensitivity of full fine-tuning to pooling is consistent with the concept becoming distributed across tokens, which, as RQ2 shows, also reduces alignment between DLA and expert words. CLEAN-FT starts from the probe's attention-pooled representation and fine-tunes the layers below the read-out, allowing it to adapt the representation while retaining the localized read-out.

RQ2: Localized training keeps evidence aligned and faithful without sacrificing detection. The evidence columns of Table 1 score the attention-pooled model of each row against the expert spans. The weakest DLA alignment on each backbone comes from fine-tuning with a randomly initialized head (Full FT on Gemma, ML-FT on Llama), the frozen probes are the most aligned, and CLEAN-FT sits between them, close to the probes on Gemma. The same ordering holds under mean pool-

Table 1: Detection and alignment on LonXplain. AUROC under mean and attention pooling, with Δ their paired difference. AUPRC and ΔAOPC on attention-pooling except for SAE, MentalBERT, and GloVe+BiGRU. †Integrated Gradients for MLP.

Type	Method	AUROC			AUPRC		ΔAOPC
		Mean	Attention	Δ	DLA	Occlusion	DLA
Gemma-2-2B							
Fine-tune	Full FT	**0.927**±.012	0.931±.012	+0.004	0.334±.026	0.495±.029	0.851±.151
	ML-FT	0.926±.012	0.931±.012	+0.005	0.398±.027	0.519±.028	1.012±.148
Hybrid	LP-FT	0.895±.015	0.929±.012	+0.035	0.375±.025	0.463±.032	1.198±.247
	CLEAN-FT	0.887±.015	**0.933**±.011	+0.046	0.491±.025	0.553±.033	1.398±.144
Probe	Linear	0.837±.019	0.930±.012	+0.092	**0.504**±.027	**0.588**±.032	1.672±.182
	MLP	0.872±.016	0.924±.012	+0.052	0.493±.027†	0.576±.032	**2.177**±.475†
	SAE	0.870±.016	–	–	0.434±.029	0.402±.028	0.729±.165
Llama-3.1-8B							
Fine-tune	Full FT	**0.918**±.013	0.920±.013	+0.002	0.236±.023	0.455±.030	0.601±.093
	ML-FT	0.911±.013	0.914±.013	+0.002	0.161±.019	0.464±.031	0.325±.105
Hybrid	LP-FT	0.877±.016	0.930±.012	+0.052	0.446±.027	**0.488**±.033	1.255±.193
	CLEAN-FT	0.865±.016	**0.931**±.012	+0.066	0.383±.026	0.472±.030	0.949±.130
Probe	Linear	0.835±.018	0.905±.014	+0.070	0.420±.027	0.440±.029	1.356±.166
	MLP	0.868±.017	0.885±.015	+0.017	**0.475**±.028†	0.445±.029	**1.623**±.277†
	SAE	0.898±.015	–	–	0.410±.028	0.405±.028	1.307±.146
Reference	Span-FT (Gemma)	0.910±.014	0.924±.013	+0.014	0.618±.026	0.458±.032	1.284±.220
	Span-FT (Llama)	0.892±.015	0.911±.013	+0.020	0.615±.027	0.442±.030	1.663±.190
	MentalBERT	0.918±.013	–	–	0.183±.019	0.537±.030	0.478±.062
	GloVe+BiGRU	0.844±.018	–	–	0.107±.015	0.407±.030	0.351±.127
Floor	Random ranking	0.5	0.5	–	0.083±.010	0.083±.010	0.0

ing (Appendix D), so the gap comes from the training method rather than the pooling strategy.

DLA and occlusion disagree where the concept has been delocalized. For Full FT, occlusion alignment stays close to that of the other methods while DLA alignment falls, suggesting that the model still depends on expert words but distributes its score across other positions. Forcing the decomposition onto the spans with span supervision (Span-FT) raises DLA alignment above every other method without improving occlusion or AUROC, while the non-LLM baselines (MentalBERT, GloVe+BiGRU) combine near-floor DLA alignment with substantial occlusion reliance. High DLA alignment alone therefore guarantees neither reliance on the evidence nor better detection.

Table 1 also separates the two localization mechanisms. Moving only the read-out to the intermediate layer (ML-FT versus Full FT) helps on Gemma but hurts on Llama, because a random head still drives coupled updates that distort the representation below the read-out. Adding a probe-initialized head (CLEAN-FT versus ML-FT) raises DLA alignment on both backbones, from 0.398 to 0.491 on Gemma and from 0.161 to 0.383 on Llama, and raises faithfulness in step. Moving a solved probe from the final to the intermediate layer (CLEAN-FT versus LP-FT) helps on Gemma but not on Llama, which suggests that loneliness sits nearer the final layer in Llama-3.1-8B than our a priori depth assumes.

Faithfulness (ΔAOPC) follows alignment rather than detection. Among the methods that read the residual stream, the probes are the most faithful, fine-tuning from a random head the least, and the hybrids fall between, whereas their detection differences are comparatively small. High AUROC can therefore mask decision logic that experts cannot check, which is why expert-aligned evidence matters for clinical auditing and trust (Doshi-Velez and Kim, 2017; Geirhos et al., 2020). In localization terms, the methods whose decomposition stays on the expert spans

Table 2: Expert alignment in and out of distribution. All methods use attention pooling. Expert annotators had 64.7% label agreement and a word-level Jaccard of 0.330.

	In-distribution LonXplain		Out-of-distribution Expert 1		Out-of-distribution Expert 2	
	AUPRC	ΔAOPC	AUPRC	ΔAOPC	AUPRC	ΔAOPC
Gemma-2-2B						
Full FT	0.334	0.851	0.346	0.772	0.260	0.865
ML-FT	0.398	1.012	0.350	1.256	0.262	1.354
LP-FT	0.375	1.198	0.394	1.971	0.301	1.520
CLEAN-FT	0.491	1.398	**0.409**	**2.844**	**0.331**	**1.974**
Linear	**0.504**	1.672	0.368	1.784	0.284	1.528
MLP	0.493	**2.177**	0.347	1.796	0.267	1.601
Llama-3.1-8B						
Full FT	0.236	0.601	0.300	0.487	0.172	0.484
ML-FT	0.161	0.325	0.298	0.475	0.157	0.572
LP-FT	0.446	1.255	0.382	1.401	0.280	1.248
CLEAN-FT	0.383	0.949	0.373	1.249	**0.297**	1.292
Linear	0.420	1.356	**0.387**	1.525	**0.297**	1.607
MLP	**0.475**	**1.623**	0.360	**1.886**	0.287	**1.960**

are the methods whose read-out stays where the concept is represented.

RQ3: Localized methods keep evidence alignment and faithfulness under distribution shift. Table 2 transfers the attention-pooled detectors to the expert-annotated conversations without retraining. Fine-tuning from a random head (Full FT, ML-FT) is the least faithful method on both backbones and the least aligned on Llama. The frozen probes stay faithful under shift, but on Gemma their alignment falls to nearly the level of Full FT, and a frozen backbone cannot adapt. CLEAN-FT keeps the concept localized while adapting the representation: on Gemma it is the most aligned and most faithful method for both annotators, above the frozen probes, and on Llama it stays close to the linear probe on alignment with lower faithfulness, consistent with the in-distribution finding that the a priori read-out depth suits Gemma better. Overall, these results suggest that localized training helps preserve evidence alignment and faithfulness under distribution shift.

5. Limitations

Our study has several limitations. First, DLA describes how the decision is represented over contextual token representations, not which input words caused it, while occlusion measures word-level dependence one word at a time and misses interactions. MLP probe attributions use Integrated Gradients and are not directly comparable to linear DLA; ΔAOPC is model-dependent, and SAE faithfulness is indirect because deletion recomputes the latents.

Second, our design only partially separates localization from training. We do not evaluate a frozen probe at the final layer or sweep read-out depth, and we do not directly measure representation changes during fine-tuning. The chosen depth may therefore not be optimal, particularly for Llama. Each configuration is trained once per fold, so intervals do not capture seed variation.

Third, the OOD set contains only 34 LLM-screened sessions from 7 of 16 participants. Two experts had 64.7% label agreement and a word-level Jaccard of 0.330, so we report their annotations separately. Finally, the task detects loneliness expression rather than person-level loneliness or clinical need, and no clinician evaluated model evidence; the models are therefore intended for expert-assisted analysis rather than clinical use.

6. Conclusion

Whether LLM loneliness detection aligns with expert evidence depends on how the model is trained and how its evidence is read. An attention-pooled probe on a frozen backbone detects loneliness about as well as full fine-tuning, and CLEAN-FT detects it best. Fine-tuning from a randomly initialized head produces the least expert-aligned and least faithful evidence, even though occlusion shows it still depends on the expert words, whereas the probe and CLEAN-FT keep their decomposition on the expert spans. Seen through concept localization, reading loneliness where the frozen model represents it and starting fine-tuning from a head aligned with that representation keeps the decision anchored to the concept, while fine-tuning from a random head at the final layer delocalizes it. The same pattern holds on expert-annotated conversations with older adults. When evidence must be read from the model itself, frozen probes or probe-initialized mid-layer fine-tuning give evidence that experts can check without post-hoc attribution.

References

Anthropic. Introducing claude opus 5, July 2026. URL https://www.anthropic.com/news/claude-opus-5. Accessed: 2026-09-08.

Varsha D Badal, Camille Nebeker, Kaoru Shinkawa, Yasunori Yamada, Kelly E Rentscher, Ho-Cheol Kim, and Ellen E Lee. Do words matter? detecting social isolation and loneliness in older adults using natural language processing. *Frontiers in psychiatry*, 12:728732, 2021.

Manuela Barreto, Jolien Van Breen, Christina Victor, Claudia Hammond, Alice Eccles, Matthew T Richins, and Pamela Qualter. Exploring the nature and variation of the stigma associated with loneliness. *Journal of Social and Personal Relationships*, 39(9):2658–2679, 2022.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

Emma Cho, Hannah Cho, George Demiris, Sean Harrison, Xiayan Ji, Ahhyun Yuh, Oleg Sokolsky, and Insup Lee. Older adults' perceptions and attitudes toward passive sensors to measure loneliness: a qualitative study. *Sage open aging*, 11: 30495334251325607, 2025.

Jenny De Jong-Gierveld and Theo G van Tilburg. A 6-item scale for overall, emotional, and social loneliness: Confirmatory tests on survey data. *Research on aging*, 28(5):582–598, 2006.

Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. Eraser: A benchmark to evaluate rationalized nlp models. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4443–4458, 2020.

Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017.

Muskan Garg, Chandni Saxena, Debabrata Samanta, and Bonnie J Dorr. Lonxplain: Lonesomeness as a consequence of mental disturbance in reddit posts. In *International conference on applications of natural language to information systems*, pages 379–390. Springer, 2023.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

Sharath Chandra Guntuku, Rachelle Schneider, Arthur Pelullo, Jami Young, Vivien Wong, Lyle Ungar, Daniel Polsky, Kevin G Volpp, and Raina Merchant. Studying expressions of loneliness in individuals using twitter: an observational study. *BMJ open*, 9(11):e030355, 2019.

Zhengfu He, Wentao Shu, Xuyang Ge, Lingjie Chen, Junxuan Wang, Yunhua Zhou, Frances Liu, Qipeng Guo, Xuanjing Huang, Zuxuan Wu, et al. Llama scope: Extracting millions of features from llama-3.1-8b with sparse autoencoders. *arXiv preprint arXiv:2410.20526*, 2024.

Julianne Holt-Lunstad, Timothy B. Smith, Mark Baker, Tyler Harris, and David Stephenson. Loneliness and social isolation as risk factors for mortality: A meta-analytic review. *Perspectives on Psychological Science*, 10:227–237, 2015. doi: 10.1177/1745691614568352.

Robert Huben, Hoagy Cunningham, Logan Smith, Aidan Ewart, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations*, volume 2024, pages 7827–7845, 2024.

Paul Jaccard. The distribution of the flora in the alpine zone. 1. *New phytologist*, 11(2):37–50, 1912.

Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable nlp systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4198–4205, 2020.

Shaoxiong Ji, Tianlin Zhang, Luna Ansari, Jie Fu, Prayag Tiwari, and Erik Cambria. Mentalbert: Publicly available pretrained language models for mental healthcare. In *proceedings of the thirteenth language resources and evaluation conference*, pages 7184–7190, 2022.

Yueyi Jiang, Yunfan Jiang, Liu Leqi, and Piotr Winkielman. Many ways to be lonely: fine-grained characterization of loneliness and its potential changes in covid-19. In *Proceedings of the*

International AAAI Conference on Web and Social Media, volume 16, pages 405–416, 2022.

Michelle Damin Kim, Ellie S Paek, Yufen Lin, Emily Mroz, Jane Chung, and Jinho D Choi. Why are we lonely? leveraging llms to measure and understand loneliness in caregivers and non-caregivers. In *Proceedings of the 1st Workshop on Linguistic Analysis for Health (HeaLing 2026)*, pages 222–235, 2026.

Polina Kirichenko, Pavel Izmailov, and Andrew Gordon Wilson. Last layer re-training is sufficient for robustness to spurious correlations. *arXiv preprint arXiv:2204.02937*, 2022.

Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. *arXiv preprint arXiv:2202.10054*, 2022.

Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300, 2024.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372, 2022.

Sam Rickman, Jose-Luis Fernandez, and Juliette Malley. Understanding patterns of loneliness in older long-term care users using natural language processing with free text case notes. *PloS one*, 20 (4):e0319745, 2025.

Daniel W Russell. Ucla loneliness scale (version 3): Reliability, validity, and factor structure. *Journal of personality assessment*, 66(1):20–40, 1996.

Wojciech Samek, Alexander Binder, Grégoire Montavon, Sebastian Lapuschkin, and Klaus-Robert Müller. Evaluating the visualization of what a deep neural network has learned. *IEEE transactions on neural networks and learning systems*, 28 (11):2660–2673, 2016.

Oscar Skean, Md Rifat Arefin, Dan Zhao, Niket Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. Layer by layer: Uncovering hidden representations in language models. *arXiv preprint arXiv:2502.02013*, 2025.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017.

Ning Wang, Sanchit Goel, Stephanie Ibrahim, Varsha D Badal, Colin Depp, Erhan Bilal, Koduvayur Subbalakshmi, and Ellen Lee. Decoding loneliness: Can explainable ai help in understanding language differences in lonely older adults? *Psychiatry research*, 339:116078, 2024.

Yasunori Yamada, Kaoru Shinkawa, Miyuki Nemoto, and Tetsuaki Arai. Automatic assessment of loneliness in older adults using speech analysis on responses to daily life questions. *Frontiers in Psychiatry*, 12:712251, 2021. doi: 10.3389/fpsyt.2021.712251.

Kailai Yang, Shaoxiong Ji, Tianlin Zhang, Qianqian Xie, Ziyan Kuang, and Sophia Ananiadou. Towards interpretable mental health analysis with large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 6056–6077, 2023.

Appendix A. Chatbot Data

The out-of-distribution corpus was collected across two deployments of a voice-based wellbeing chatbot with community-dwelling older adults between December 2025 and August 2026. 16 participants held 383 sessions.

The chatbot in Figure 4 was deployed on a Samsung tablet in each participant's home, where it collected a diverse range of conversations, including scheduled daily mood check-ins and open-ended sessions in which the model drew on stored memories of earlier conversations to ask follow-up questions. The tablet application runs a real-time voice pipeline combining speech-to-text, voice activity detection, and text-to-speech, so that a session proceeds as continuous spoken dialogue rather than typed turns. At installation, each participant reviewed a consent form and completed a profile (preferred name, age, household members, city, and free-text interests) that was supplied to the model as part of its system prompt. Check-in sessions were prompted by push notifications on the tablet to remind participants to use the chatbot. Participants were instructed to interact with the chatbot daily and were free to start multiple sessions per day. Transcripts, timestamps, and conversation IDs were logged and stored securely on institutional servers.

A management dashboard shown in Figure 5 gives the study team oversight of the deployment without altering the conversations themselves. It lists sessions with their start and end times, message counts, and full conversation logs with speaker labels, and it allows the team to configure daily check-in notifications and reminders for participants who did not respond. A guardrail layer screens user input in real time and blocks unsafe requests, including content related to suicidal ideation, before they reach the model; flagged sessions are highlighted on the dashboard for prompt review and the study team is notified. Both mechanisms were a condition of IRB approval for deployment with older adults.

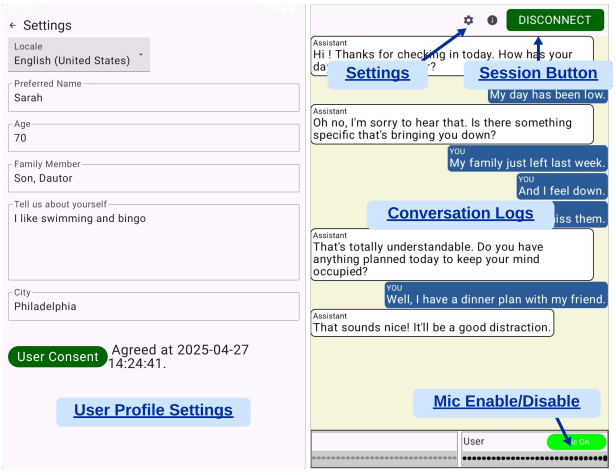


Figure 4: Screenshots of the Android application. The left panel shows the user profile settings, and right panels show the main home screen for conversation.

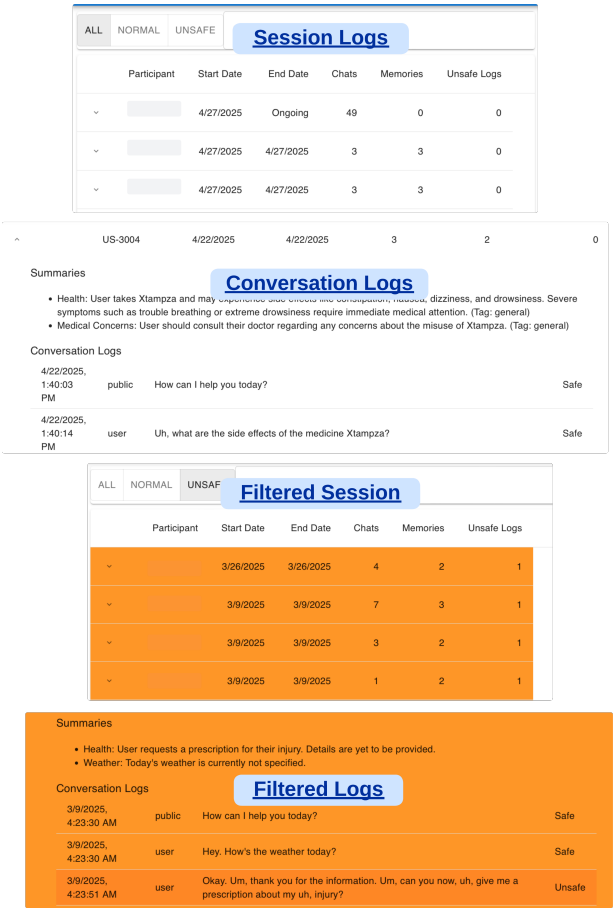


Figure 5: Management dashboard for the study team. The dashboard records user interactions and highlights sessions flagged as unsafe by the guardrail.

Appendix B. Screening Prompt

```
screening_prompt: |-
  CLINICAL SCREENING DEFINITION OF LONESOMENESS

  Lonesomeness is an unpleasant emotional response to
  perceived isolation through mind and character,
  especially in terms of concerns for social lifestyle.

  You are an expert clinical psychologist screening spoken
  conversations between older adults and a wellbeing
  chatbot. Read the exchange below and decide whether the
  older adult's reply expresses lonesomeness under the
  screening definition above.

screening_output: |-
  Answer 1 if the older adult expresses lonesomeness
  anywhere in this exchange, and 0 if not. Reply with that
  single digit and nothing else.
```

Appendix C. Annotation Rubric

Clinical Definition of Lonesomeness

Lonesomeness is an unpleasant emotional response to perceived isolation through mind and character, especially in terms of concerns for social lifestyle.

To mark lonesomeness as present, annotators must identify at least one of its three core dimensions:

- **Social loneliness:** The perceived absence of a supportive social network or group.
- **Emotional loneliness:** The perceived absence of a close, intimate partner, companion, or relationship.
- **Existential loneliness:** A deep, subjective feeling of disconnection from the broader world or humanity.

Strict Annotation Rules for Difficult Cases

Lonesomeness in the Past. A person with a history of loneliness may still be at risk of self-harm or suicide. For instance:

> *"I was so upset being lonely before Christmas and today I am celebrating New Year with friends."*

We define rules to capture indicators of prior lonesomeness, such as attending a celebration to fill the void associated with this negative emotion. This post is thus marked as presenting lonesomeness, functioning as an indicator that the author is potentially at risk.

Ambiguity with Social Lonesomeness. Major interpersonal events (e.g., breakups, marriage, conflicts with best friends) may be mentioned in different contexts, suggesting distinct perceptions. We formulate two specific annotation rules:

- **Negative affect / longing:** Any expression of a void, missing someone, regret, or mentioning such events with negative valence is marked as the **presence** of lonesomeness.

  > *"But I just miss her SO. much. It's like she set the bar so high that all I can do is just stare at it."*
  > → **LONELY**

- **General conflict / situational stress:** Mentions of fights, quarrels, or general stories without explicit expressions of emotional isolation are marked as the **absence** of lonesomeness.

  > *"My husband and I just had a huge argument and he stormed out. I should be crying or stopping him or something. But I decided to take a handful of benzos instead."*
  > → **NOT LONELY**

Appendix D. Mean-pooled Results

Table 3 shows evidence alignment metrics (AUPRC DLA and ΔAOPC) for mean-pooled methods. AUROC is shown in Table 1.

Table 3: Evidence alignment for mean-pooled models on LonXplain. †Integrated Gradients for MLP.

Type	Method	Gemma-2-2B		Llama-3.1-8B	
		AUPRC	ΔAOPC	AUPRC	ΔAOPC
Fine-tune	Full FT	0.307	0.846	0.227	0.741
	ML-FT	0.392	0.945	0.153	0.270
Hybrid	LP-FT	0.434	0.869	0.395	1.258
	CLEAN-FT	0.459	**1.609**	0.415	**1.472**
Probe	Linear	0.416	1.372	0.404	1.168
	MLP	**0.461**†	0.788†	**0.433**†	0.471†
	SAE	0.434	0.729	0.410	1.307